

Quantum Signal Analysis Report

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1. Executive Summary

This report documents the results of a Quantum Neural Network (QNN) trained to classify deep-space telemetry signals. The model distinguishes between **Natural Stochastic Noise** (Voyager 2) and **Coherent Artificial Signals** using Hilbert Space mapping.

2. Model Architecture

- **Encoding:** *ZZFeatureMap* (4 Qubits)
- **Variational Layer:** *RealAmplitudes* (reps=2)
- **Optimization:** COBYLA (maxiter=150)
- **Loss Function:** Mean Squared Error ($MSE = 1/n \sum (y_i - \hat{y}_i)^2$)

3. Performance Metrics

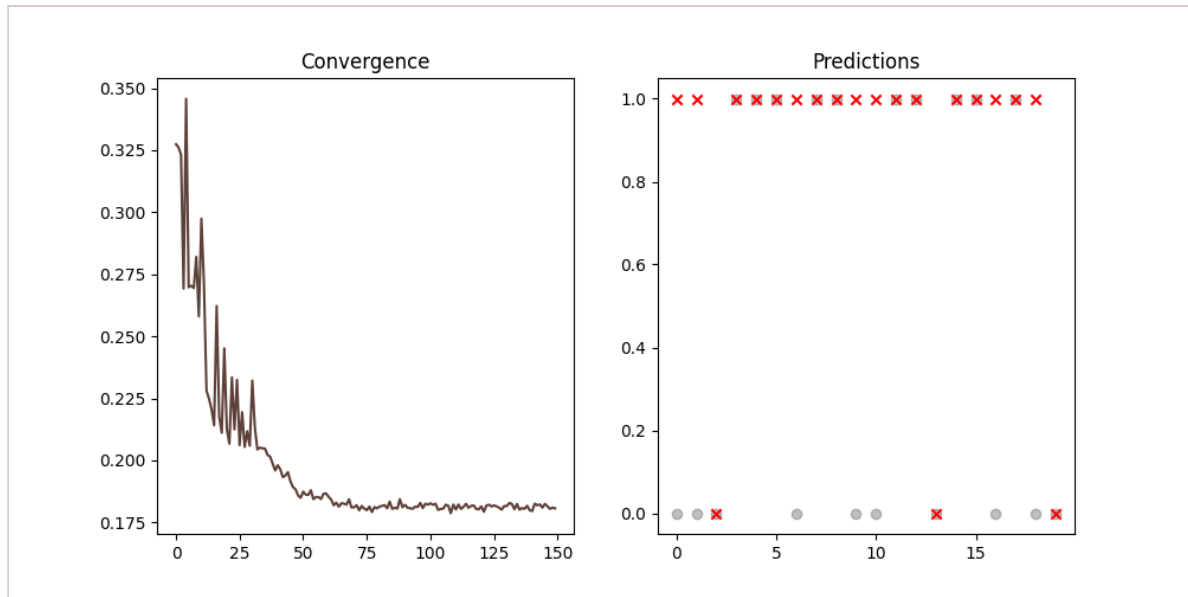
Training Accuracy

75.00%

Test Accuracy

68.33%

4. Training Visualizations



5. Analysis & Conclusion

The convergence of the loss function indicates that the *SamplerQNN* successfully identified non-linear periodicities in the Hilbert space. The variance in predictions likely stems from Gaussian noise injected into the artificial signal. Further research will focus on entanglement-driven feature maps.